

Résumé—Job Matching & Talent Marketplace Engine
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Résumé–Job Matching & Talent Marketplace Engine** is an intelligent recruitment platform designed to improve the efficiency and accuracy of the hiring process using advanced Data Structures and Algorithms (DSA). The system indexes résumés and job postings, normalizes inconsistent skill names, and compares candidate profiles with job requirements to calculate a compatibility score. Based on this score, candidates are ranked and suitable job recommendations are generated. The platform also performs optimal candidate-to-role assignment, ensuring the best candidates are selected for available positions while maximizing overall matching quality. Additionally, it identifies the minimum set of skills and candidates required to form an effective project team, helping organizations utilize resources efficiently. The project incorporates **String Algorithms** for efficient skill extraction and matching, **Advanced Dynamic Programming** for similarity computation and ranking, **Network Flow and Bipartite Matching** algorithms for optimal candidate-to-role assignment, **Approximation Algorithms** for determining the minimum skill set required for team formation, and **Randomized Hashing** for fast indexing and scalable search operations. This solution reduces manual effort, improves recruitment accuracy, and offers a scalable approach for modern hiring platforms handling large volumes of applicants and job opportunities.

Signature of the Guide